Department of Computer Science & Engineering

Software Requirement Specification

Course Code – CSE 3411

Course Title – System Analysis & Design

Project Title – EmpowerHub

Submitted To

Didarul Islam Didar

Lecturer, Dept. of CSE, UIU

Submitted By

Team Incredible

Afnan Shahriar – 011213001

Md. Habibullah – 011221260

Tamanna Islam – 011221318

Md. Jana Alam – 011221353

Nazmul Alam – 0112230914

Khondokar Marzia – 0112310464

Date of Submission – 16th September, 2025

Abstract

EmpowerHub is an innovative e-learning platform aimed at redefining the learning experience by combining education, collaboration, and well-being. The platform allows learners to acquire diverse skills from professional instructors while also engaging in peer-to-peer knowledge exchange through a unique credit-swapping system. Beyond skill development, EmpowerHub addresses the vital aspect of mental health by providing access to professional counseling services. It integrates a variety of features such as course offerings, online software tools, skill matchmaking, nearby instructor discovery, community forums, review and rating mechanisms, and a reward badge system to foster active participation. The platform supports five interconnected roles—students, instructors, organizations, counselors, and administrators—working collaboratively to create a holistic and engaging learning ecosystem.

Acknowledgement

We would like to express our deepest gratitude to all those who supported and guided us throughout the development of EmpowerHub. Our sincere thanks go to our respected faculty members for their valuable advice, encouragement and constructive feedback during each stage of this project. We are also grateful to our peers and friends who shared insightful suggestions and inspired us to refine our ideas. Finally, we acknowledge the collaborative effort of our team members, whose dedication, creativity and commitment made this project possible.

1 Introduction

1. 1 Overview

EmpowerHub is an innovative e-learning platform designed to empower learners with the opportunity to acquire diverse skills from experienced instructors. In addition to learning, the platform fosters collaboration by enabling learners to exchange their skills with others through a unique credit-swapping system. Recognizing the importance of mental health in the learning journey, EmpowerHub also provides access to mental counseling services from professional counselors.

The platform offers a rich variety of courses, online software products, skill matchmaking, and finding nearest instructors, forums, review-rating mechanisms, and a reward badge system to encourage engagement and recognize achievements. Five distinct roles work collaboratively to drive the platform: students, instructors, organizations, counselors, and administrators.

1.2 Problem Statement

Modern e-learning platforms often focus solely on content delivery, overlooking critical aspects such as community collaboration, learner mental health, and personalized engagement. Traditional models lack effective mechanisms for learners to monetize their skills or gain recognition for their contributions. Additionally, the absence of integrated mental health support can leave learners feeling isolated or demotivated during their educational journey.

1.3 Objective

The primary objective of EmpowerHub is to create a holistic and inclusive e-learning ecosystem where:

- Learners can develop and skill exchange
- Mental health support is seamlessly integrated
- A community driven environment encourages collaboration, growth and recognition.
- Organizations can upskill their workforce efficiently.

1.4 Motivation

The motivation behind EmpowerHub stems from the growing demand for personalized, community-driven learning solutions that cater to both skill development and emotional well-being. By addressing these needs, EmpowerHub aims to stand out as a transformative platform, bridging gaps in the e-learning space while fostering a sense of belonging and achievement for its users.

2 System Development Life Cycle (SDLC)

The development of EmpowerHub follows the System Development Life Cycle (SDLC), ensuring a structured and systematic approach to building the platform. The steps are outlined below:

2.1 Planning

In this phase, the project scope, objectives, and resources were identified. The primary goal was to design an e-learning platform that integrates skill development, collaboration and mental health support. Stakeholders, including students, instructors, organizations, counselors and administrators, were identified as the core users of the system.

2.2 System Information

Detailed requirements were collected through surveys, interviews and document reviews. Both functional (course enrollment, credit-swapping, counseling booking, forums, reward system) and non-functional requirements (security, scalability, user-friendliness) were gathered.

2.3 Analysis

The collected requirements were analyzed to identify gaps in existing e-learning platforms and to ensure that EmpowerHub addresses critical needs such as collaboration, recognition and mental health integration. Feasibility in terms of technology, finance and user adoption was also assessed.

2.4 Design

The system design phase involved creating architectural blueprints and interface mockups. The design covered:

- Database schema using MySQL.
- Front-end interface with HTML, CSS, Tailwind and Bootstrap for responsiveness.
- Backend workflows using PHP for server-side logic.
- User roles and permissions (students, instructors, organizations, counselors, administrators).

2.5 Development/Coding

The system was developed using the chosen technologies. Frontend modules (course pages, forums, dashboards) and backend modules (credit-swapping engine, counseling scheduling, admin panel) were coded in alignment with the design documents.

2.6 Testing

Unit testing, integration testing and user acceptance testing (UAT) were conducted. Test cases ensured correct implementation of functionalities like course enrollment,

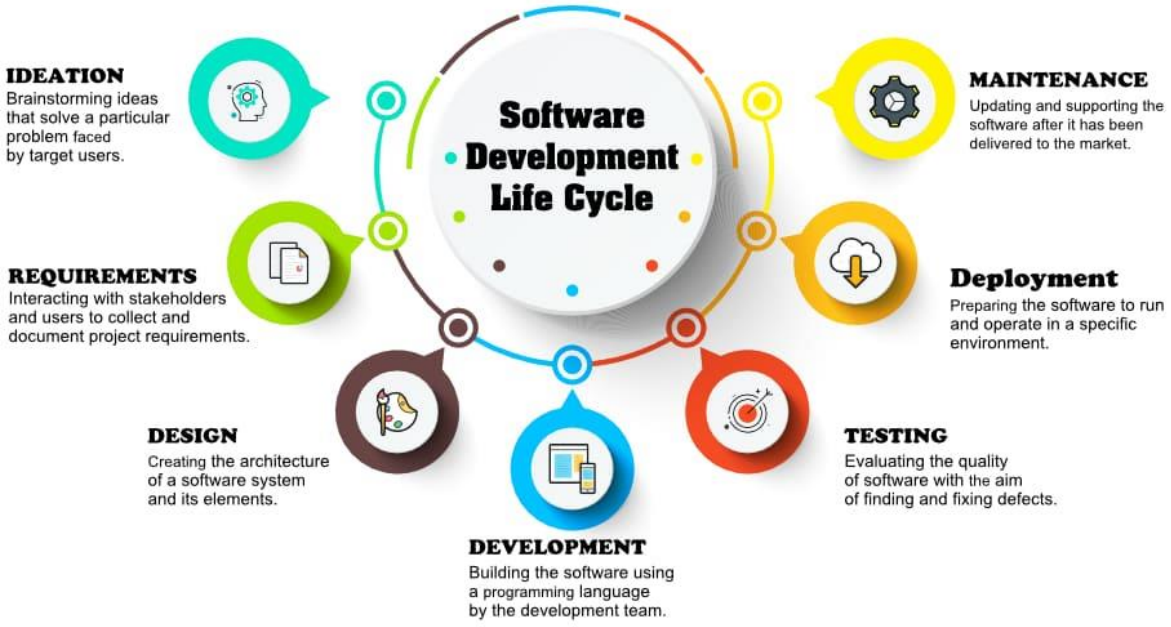
credit transactions, counseling booking and review-rating mechanisms. Security and performance testing were also prioritized.

2.7 Deployment

The platform was deployed on a secure hosting server, making it accessible to the intended user groups. Necessary configurations for scalability, load balancing and security were set up during this stage.

2.8 Post-implementation & Maintenance

Continuous monitoring and maintenance ensure bug fixes, system updates and new features to make enhancements. User feedback loops (via surveys and forums) are established to refine the system and improve user experience over time.



3 Requirement Analysis

3.1 Vision

To become the leading e-learning platform that not only empowers individuals with skills but also fosters a collaborative, supportive, and mentally enriching learning environment.

3.2 Mission

- Providing high quality, diverse courses led by experienced instructors.
- Introducing innovative mechanisms like credit swapping for skill exchange.
- Integrating mental health support as a core component of the learning process.
- Encouraging community engagement through forums, reviews, and rewards.
- Ensuring seamless interaction and coordination among students, instructors, organizations, counselors, and administrators.

3.3 Goals

- Develop a user-friendly platform with robust features such as course enrollment, credit swapping, and mental health appointment scheduling.
- Provide tools for instructors to create and manage engaging courses.
- Implement reward mechanisms to enhance user motivation.
- Build a secure and scalable system capable of handling large volumes of users and data.
- Foster a supportive community through active forums and transparent review systems.

3.4 Information Sources

To gather requirements for EmpowerHub, only external information sources were used.

- ✓ **External Sources:** Market research reports, literature reviews on e-learning trends, competitor platform analysis and professional counselor networks.

3.5 Information Gathering Tools

The following tools and techniques were used to gather system requirements:

- **Review of Documents, Procedures and Forms** – Analyzing existing e-learning systems and academic records.
- **Interviews** – Conducting discussions with students, instructors and counselors to identify their needs.
- **Questionnaires** – Distributing surveys to potential users for broader input.
- **Online Sources** – Benchmarking existing platforms, studying research papers and reviewing forums/blogs for best practices.

3.6 Benchmark Analysis

For the benchmark study we collected data from 9 similar platforms. These platforms are: Skillshare, Udemy, Coursera, LinkedIn Learning, BetterHelp, Calm, Headspace, Talkspace, Duolingo

Following is our conducted benchmark analysis:

Feature	Skill Matchmaking	Credit Swapping	Video Session	Feedback & Rating	Skill Tracking & Rewards	Discussion Groups	Counseling Session	Workshops
Skillshare	✗	✗	✗	✗	✓	✗	✗	✗
Udemy	✗	✗	✗	✓	✓	✗	✗	✗
Coursera	✗	✗	✓	✓	✓	✗	✗	✗
LinkedIn Learning	✗	✗	✗	✓	✓	✗	✗	✗
BetterHelp	✗	✗	✓	✗	✗	✗	✓	✗
Calm	✗	✗	✗	✗	✗	✗	✗	✗
Headspace	✗	✗	✗	✗	✗	✗	✗	✗
Talkspace	✗	✗	✓	✗	✗	✗	✓	✗
Duolingo	✗	✗	✗	✗	✓	✗	✗	✗
EmpowerHUB	✓	✓	✓	✓	✓	✓	✓	✓

3.6 Gap Analysis

From the above benchmark analysis, we have encountered some of the common problems that users usually face during in a learning process. In this segment, we have described all of the gaps and show some solutions to remove the gaps as well.

3.6.1 Current Challenges

- Limited opportunities for learners to exchange their skills or gain recognition.
- Lack of integrated mental health support in traditional e-learning platforms.
- Ineffective reward systems that fail to motivate learners or track progress effectively.
- Minimal collaboration among various stakeholders (students, instructors, organizations, counselors, and admin).

3.6.2 Desired State

- A platform where learners can not only gain skills but also exchange them seamlessly.
- Integrated mental health services accessible to all learners.
- Comprehensive reward and recognition systems to celebrate milestones and contributions.
- Strong community engagement through forums and other interactive tools.

3.6.3 Approach to Bridge the Gap

- Design and implement a credit-swapping mechanism to enable skill exchange.
- Partner with certified mental health counselors for seamless appointment scheduling.
- Develop an intuitive interface with robust functionalities for each user role.
- Introduce inspiring elements like reward badges and to encourage participation and excellence.
- Build a scalable infrastructure to support a growing user base while maintaining data security and performance.

3.7 Feature List Fixation

After the benchmark and gap analysis, we have decided to fix the following features:

- **Skill Matchmaking:** Allow users to create profiles showcasing skills
- **Discussion Groups:** Community focused group discussion
- **Credit Swap:** Balanced exchange of learning and teaching
- **Feedback & Rating:** After each exchange, users can provide feedback
- **Skill Progress & Rewards:** Motivate users with badges & milestones
- **Finding Instructors:** Students can find instructors by matching their provided locations
- **Counseling Session:** Mental wellness check-ins, resources & suggestions

4 Feasibility Analysis

The feasibility of EmpowerHub was examined in multiple aspects to ensure successful implementation:

4.1 Technical Feasibility

Our web project is built with the cool quartet - HTML, CSS, Tailwind and JavaScript, spiced up with Bootstrap for that extra style and responsiveness. On the back end, we've got PHP for specific tasks. We use MySQL as our database.

4.2 Economic/Financial Feasibility

The project is cost-effective as it uses open-source technologies like HTML, CSS, Tailwind, Bootstrap, PHP and MySQL, reducing software licensing costs. Hosting and maintenance expenses are manageable within projected budgets.

4.3 Human Resources Feasibility

The project team possesses the necessary expertise in web technologies, database management and UI/UX design. Additional support from mental health professionals and instructors ensures holistic system functionality.

4.4 Behavioral/Market Feasibility

There is strong demand for platforms that blend skill development with mental health support. Current trends in online learning show increased interest in community-driven platforms, ensuring high adoption potential.

4.5 Legal/Political Feasibility

The system complies with copyright laws, privacy regulations and user data protection standards. Counseling services are aligned with professional ethics and legal guidelines for mental health support.

4.6 Schedule Feasibility

The project timeline is realistic, with SDLC phases mapped out and tasks distributed among team members. Agile methods ensure timely progress and adaptability to changes.

5 SWOT Analysis

Here is the detailed strength, weakness, opportunities and threats that can be faced to implement the project by the time.

5.1 Strengths

- Unique credit-swapping system for skill exchange.
- Integration of mental health support for learners.
- Diverse roles ensuring comprehensive functionality.
- Reward and recognition mechanisms enhancing engagement.

5.2 Weakness

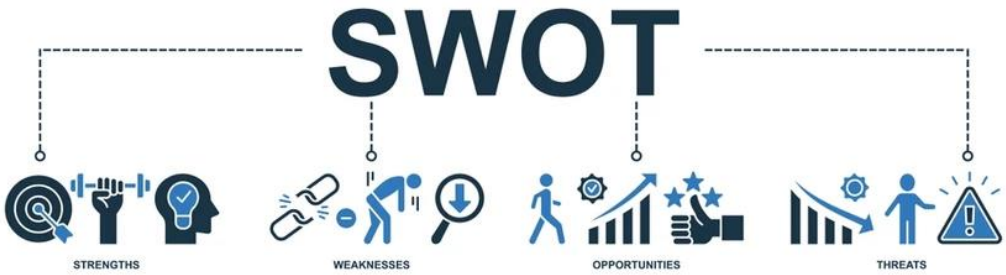
- Initial development may be resource-intensive.
- Requires high-quality mental health professionals and course instructors.
- Dependence on user activity for community-building features.

5.3 Opportunities

- Growing demand for personalized e-learning platforms.
- Potential partnerships with organizations for workforce upskilling.
- Expansion into niche skill areas and global markets.

5.4 Threats

- Competition from established e-learning platforms.
- Privacy and security concerns for user data.
- Reliance on user retention and engagement for success.



6 Cash Flow

Following is the breakdown of our revenue and expenses and then our cash flow analysis (in $dollar) for 2 years, where each column represents a quarter of a year, i.e. 3 months.

6.1 Revenue Calculation

The major revenue heads that we considered are:

- Course Subscriptions (individual learners)
- NGO Partnerships (bulk course purchases for their beneficiaries)
- Government Funding (education grants and skill-building initiatives)
- Advertisements (promotional slots on the platform)

Quarters	Year 1				Year 2			
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Course Subscriptions	15k	18k	22k	30k	35k	40k	50k	60k
NGO Partnerships	10k	12k	15k	20k	25k	30k	35k	40k
Government Funding	25k	25k	25k	25k	25k	25k	25k	25k
Advertisement	5k	7k	8k	10k	12k	15k	18k	20k
Total Revenue	**55k**	**62k**	**70k**	**85k**	**97k**	**110k**	**128k**	**145k**

6.2 Expense Calculation

Likewise, major expense heads are mentioned below:

- Development Costs (initial platform setup, feature upgrades)
- Marketing (social media ads, outreach campaigns, partnerships)
- Instructor Fees (course content creation and delivery)
- Hosting & Operations (server costs, IT support)
- Administrative Costs (staff salaries, office expenses)

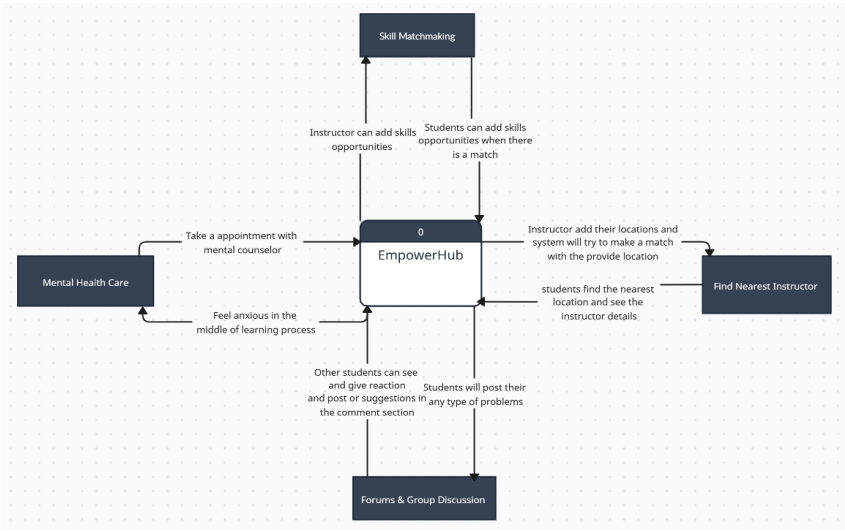
Quarters	Year 1				Year 2			
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Development Cost	20k	15k	10k	5k	5k	5k	5k	5k
Instructor Fees	8k	10k	12k	15k	18k	20k	25k	30k
Marketing	5k	6k	8k	10k	12k	15k	18k	20k
Hosting & Operations	5k	6k	7k	8k	9k	10k	11k	12k
Administrative Costs	10k	10k	12k	12k	12k	12k	12k	12k
Total Expense	**48k**	**47k**	**49k**	**50k**	**56k**	**62k**	**71k**	**79k**

6.3 Cash Flow Analysis

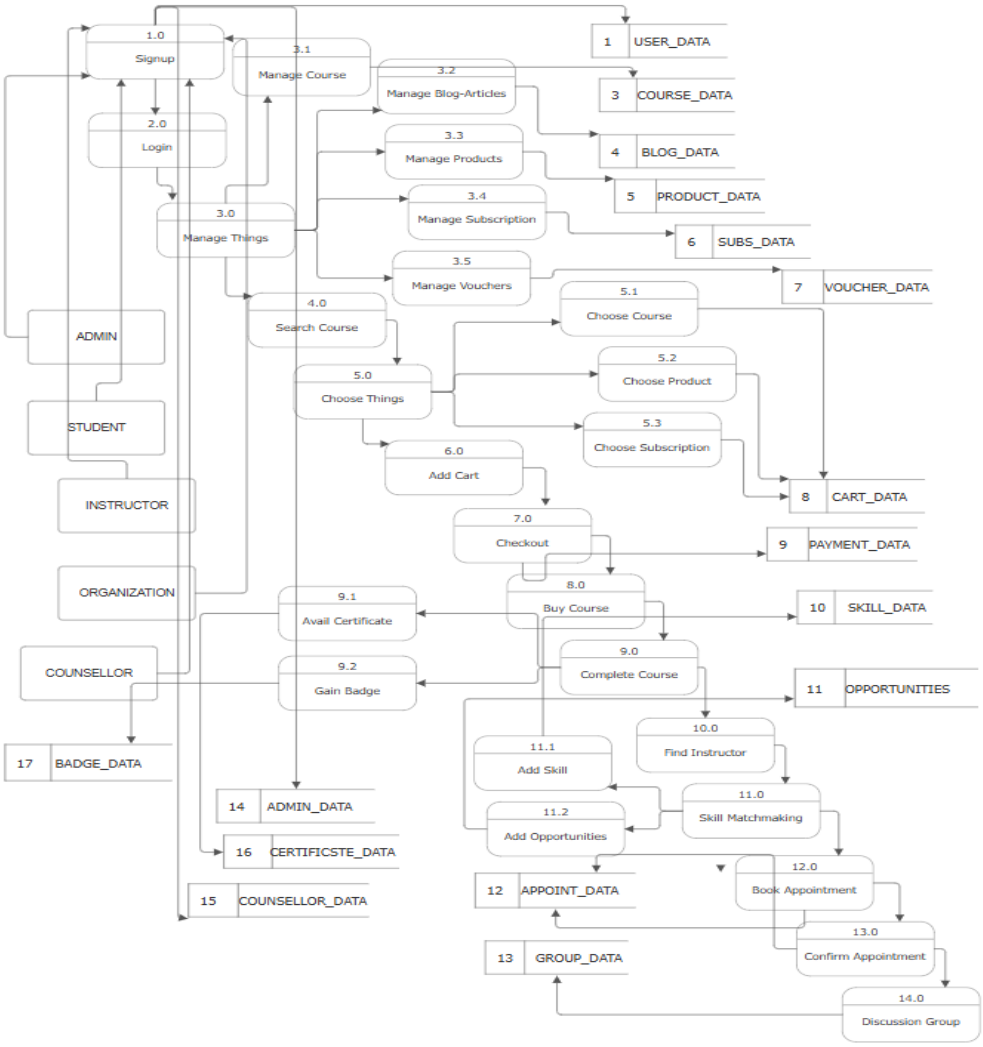
Quarters	Year 1				Year 2			
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Total Revenue	55k	62k	70k	85k	97k	110k	128k	145k
Total Expense	48k	47k	49k	50k	56k	62k	71k	79k
Net Cash Flow	**7k**	**15k**	**21k**	**35k**	**41k**	**48k**	**57k**	**66k**
Cumulative Cash	**7k**	**22k**	**43k**	**78k**	**119k**	**167k**	**224k**	**290k**

7 System Design

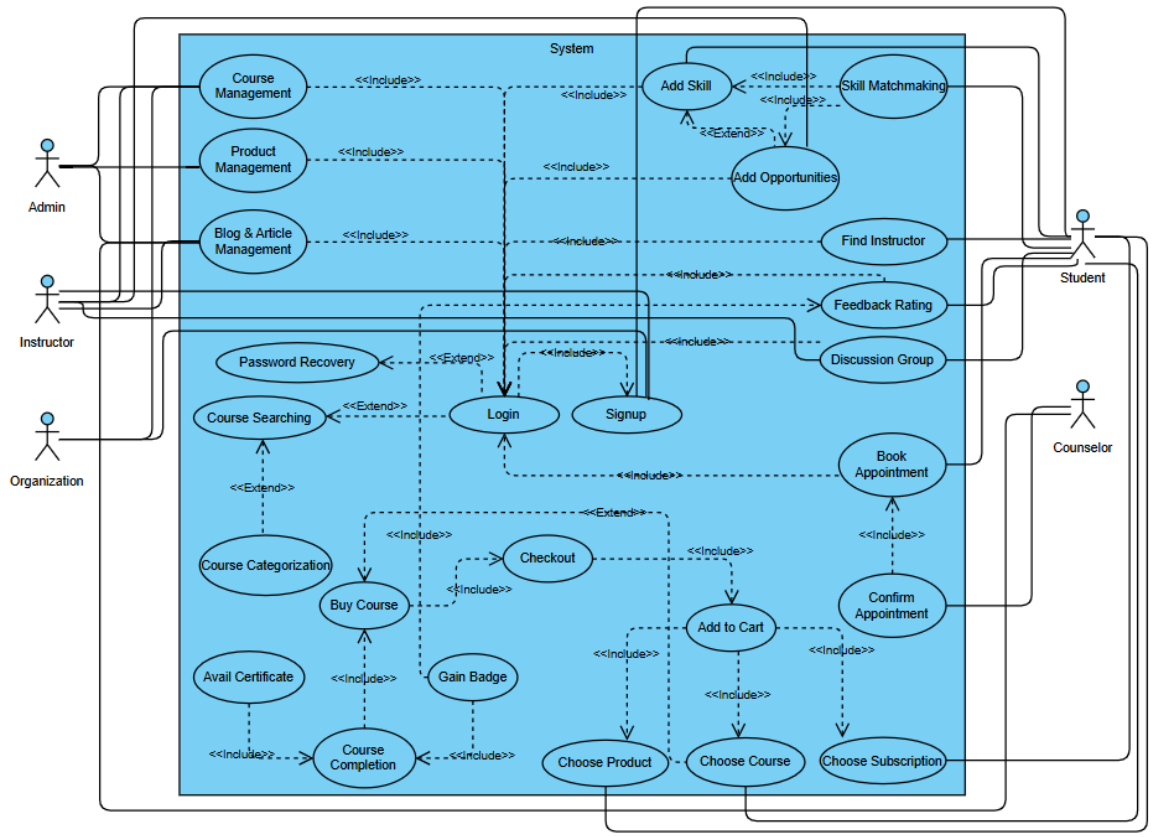
7.1 Context Diagram



7.2 Data Flow Diagram



7.3 Use Case Diagram



7.4 Use Case Descriptive Form

Use Case 1: Skill Matchmaking

Description: Student can express their learning interest and the system will make matches with the skilled instructors' opportunities.

Primary Actor: Student

Stakeholder & Interest: *Student* will add their skill and *instructor* will add their proficiency as an opportunity.

Pre-condition: Both student and instructor will have to create an account with corresponding role. At least 1 student and 1 instructor will need to be available.

Main Success Scenario: Student will navigate to 'Add Skills' in the dashboard and add his skills (what he/she know). Likewise, instructor will navigate to 'Add Opportunities' in the dashboard and add his skills as an opportunity to the student with specific pre-requirements. When there will be a match between these two objects, student will see the instructor profile & contact info in the 'Opportunities' section.

Alternative Scenario: At any time system can fail. For that, user can refresh the page and request for the previous state. If no skills can fill up the requested

opportunities, system will show a message 'No opportunities available at this moment.' Users have to wait for a time being to match with the skill exchanging.

Post-condition: The student's skills are successfully matched with an instructor's opportunities, and the student can view the instructor's profile and contact information in the 'Opportunities' section. Both student and instructor are notified about the successful match.

Use Case 2: Finding Nearest Instructor

Primary Actor: Student

Stakeholder & Interest: *Student* & *Instructor* will update their location.

Pre-condition: Both student and instructor will have to create an account with corresponding role. At least 1 student and 1 instructor will need to be available.

Main Success Scenario: Student will navigate to the dashboard and update his location. The instructor will also update his location as well. When there will be a match between these two locations, student will see the instructor profile & contact info in the 'Find Nearest Instructors' section.

Alternative Scenario: At any time system can fail. For that, user can refresh the page and request for the previous state. If no location gets matched, system will show a message 'No instructors are available nearby your location at this moment.' Users have to wait for a time being to match with the provided location.

Post-condition: The student's location is successfully matched with an instructor's location. The student is able to view the instructor's profile and contact information in the 'Find Nearest Instructors' section. Both student and instructor are notified of the successful location-based match.

Use Case 3: Forums & Group Discussion

Primary Actor: Student

Stakeholder & Interest: *Student* will post their any type of problems in the forum wall. Other students will see and give suggestions to get rid of the problems.

Pre-condition: Student will have to create an account with corresponding role. At least 2 students will need to be available.

Main Success Scenario: Student will navigate to 'My Forum' in the dashboard and post his problem that he is facing. Also he can see his own post in the main web forum wall. Other students can see his post, they can react and comment.

Alternative Scenario: At any time system can fail. For that, user can refresh the page and request for the previous state. If no post found, system will show a message 'No posts are available at this moment.

Post-condition: The problem and all associated suggestions are stored in the forum for future reference. The student gains insights or a solution to their problem. Active participation enhances community engagement within the platform.

Use Case 4: Counseling Support

Description: Student can book an appointment with a professional counselor to discuss personal, academic, or emotional challenges and receive guidance.

Primary Actor: Student

Stakeholder & Interest: Student seeks mental health support for stress, anxiety, or personal challenges. On the other hand, counselor provides professional counseling sessions to help the student manage well-being.

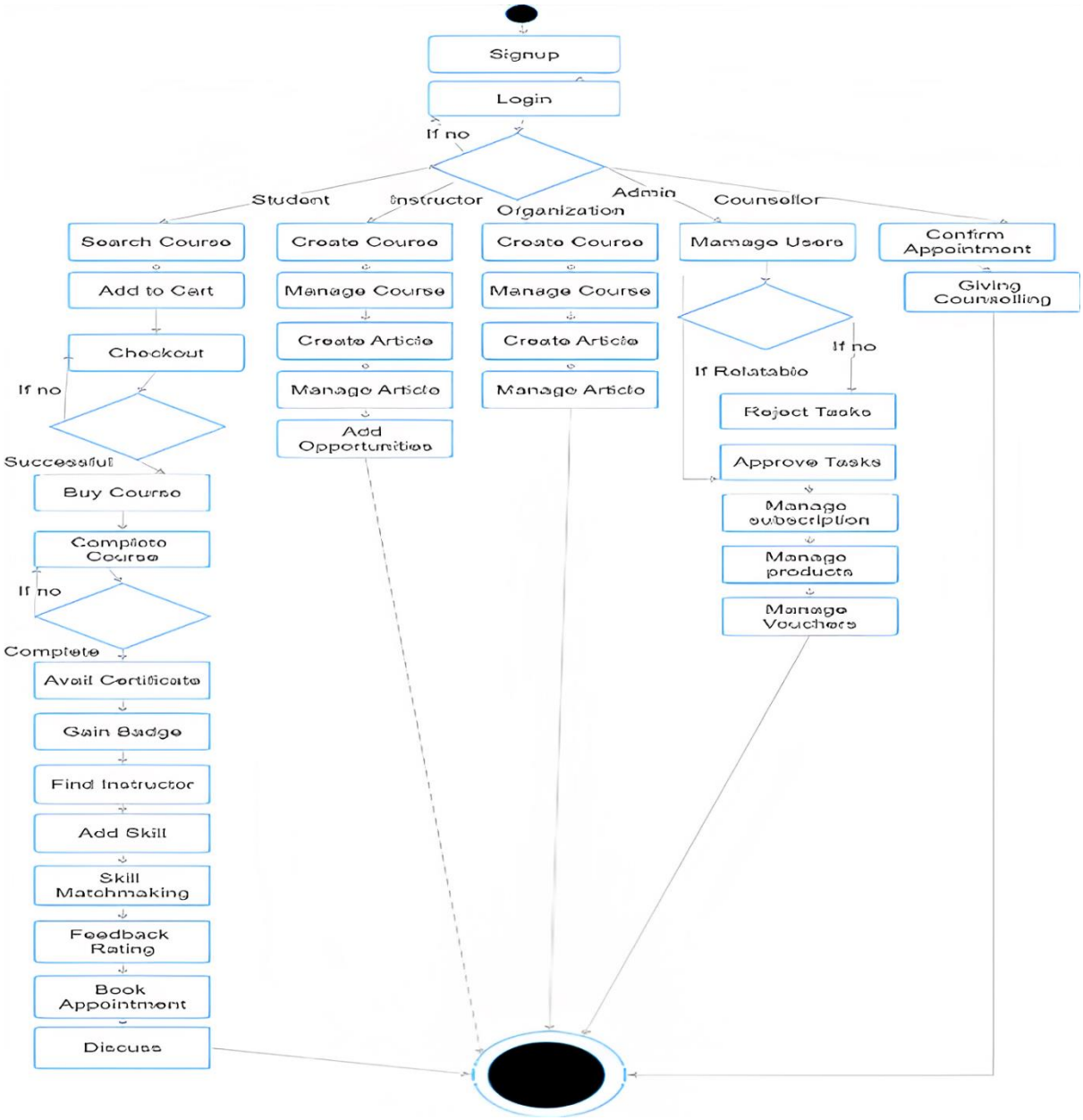
Pre-condition: Both student and counselor must have an account with their respective roles. But at least one counselor must be available and active in the system.

Main Success Scenario: Student navigates to the "Counseling" section in the dashboard viewing the list of available counselors, including their specialization and availability. Student selects a counselor and requests an appointment by choosing a suitable time slot. The counselor receives the request, reviews availability and confirms the appointment. Student receives confirmation and attends the counseling session (online or offline depending on the platform's setup).

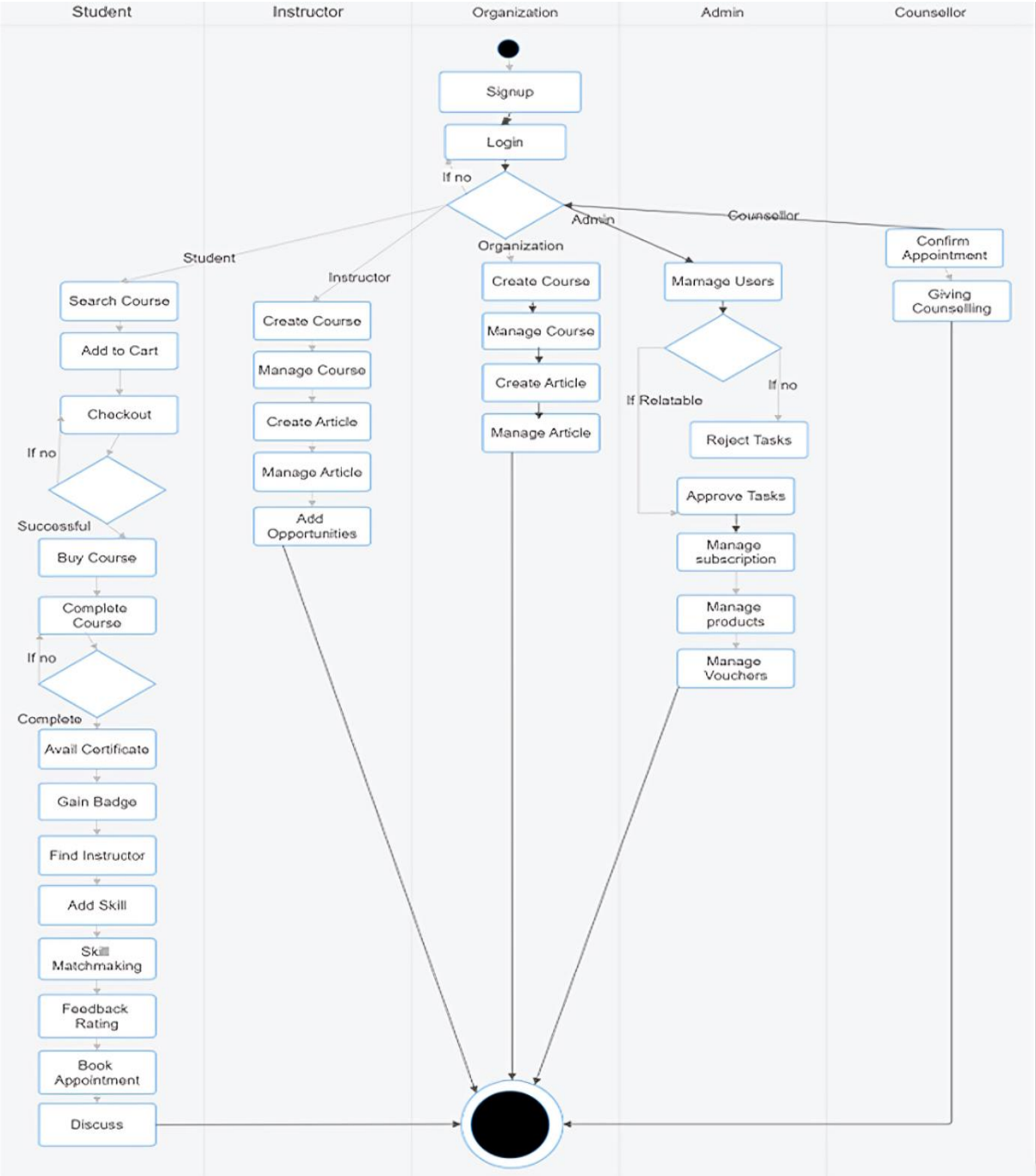
Alternative Scenario: If no counselors are available at the selected time, the system shows: "No counselors are available at this moment. Please try again later." If system failure occurs, users can refresh and retrieve their previous state. If the counselor cancels, the student is notified and asked to reschedule.

Post-conditions: The counseling appointment is successfully booked and recorded in the system. Both student and counselor are notified of the scheduled session. Session details are stored for future follow-up or reference.

7.5 Activity Diagram



7.6 Swimlane Diagram



----------The End----------